MATHEMATICAL ANALYSIS 1
HOMEWORK 8

(1) **Invertibility of continuous functions.**
   (a) Let $f(x) = x^3 + x$. Show that f is strictly increasing on $\mathbb{R}$ and that $f(\mathbb{R}) = \mathbb{R}$. Conclude that f has an inverse $f^{-1} : \mathbb{R} \to \mathbb{R}$, and that f^{-1} is continuous.
   (b) Find $f^{-1}(2)$ and $f^{-1}(-10)$.
   (c) Let $g : [0, 2] \to \mathbb{R}$ be defined by $g(x) = x$ for $0 \le x < 1$ and $g(x) = 3 - x$ for $1 \le x \le 2$. Show that g is injective but not monotone on $[0, 2]$. Why does this not contradict Theorem 9.7?

(2) **What does δ depend on?** For each of the following functions and each x_0 in the given interval I, find, for every $\varepsilon > 0$, a $\delta > 0$ such that

$$x \in I,\ |x - x_0| < \delta \quad \Longrightarrow \quad |f(x) - f(x_0)| < \varepsilon.$$

   In each case, say whether your δ depends on ε only, or also on x_0.
   (a) $f(x) = 3x - 1$, $I = \mathbb{R}$.
   (b) $f(x) = x^2$, $I = \mathbb{R}$. Then show that f is not uniformly continuous on $\mathbb{R}$ (*Hint:* consider $x_n = n$ and $y_n = n + \frac{1}{n}$), but that it is Lipschitz on $[-R, R]$ for every $R > 0$. What is a Lipschitz constant there, and which δ works on $[-R, R]$?
   (c) $f(x) = \sqrt{x}$, $I = [0, +\infty)$. (*Hint:* first show that $|\sqrt{x} - \sqrt{y}| \le \sqrt{|x - y|}$ for all $x, y \ge 0$.) Is f uniformly continuous on $[0, +\infty)$? Is it Lipschitz there?
   (d) Summarize your findings in a table: for each of (a)–(c), is f continuous, uniformly continuous, Lipschitz on I?

(3) **The absolute value is Lipschitz.** You may use the triangle inequality $|a + b| \le |a| + |b|$.
   (a) Prove the *reverse triangle inequality*: for all $x, y \in \mathbb{R}$,

$$\big||x| - |y|\big| \le |x - y|.$$

   (*Hint:* write $x = (x - y) + y$ to show $|x| - |y| \le |x - y|$, then exchange x and y.)
   (b) Deduce that $f(x) = |x|$ is Lipschitz on $\mathbb{R}$, and show that its Lipschitz constant is exactly 1. Conclude with Proposition 9.9 and Corollary 9.10 that $|x|$ is continuous and uniformly continuous on $\mathbb{R}$. Which δ works for a given ε?
   (c) Let I be an interval and $f : I \to \mathbb{R}$. Show: (i) if f is continuous at $x_0 \in I$, then so is $|f|$; (ii) if f is Lipschitz on I with constant L, then so is $|f|$. (iii) Is the converse of (i) true?
   (d) A full-wave rectifier turns an input voltage $V(t)$ into the output voltage $|V(t)|$. If the input is measured with an error of at most η, how large can the error in the output be?

(4) **Derivatives from the definition.**
   (a) Using $\lim_{y \to 0} \frac{\ln(1+y)}{y} = 1$, show that $\ln x$ is differentiable at every $x_0 > 0$, with $(\ln)'(x_0) = \frac{1}{x_0}$.
   (b) Show that $f(x) = x|x|$ is differentiable at 0, and find $f'(0)$.
   (c) Let $f(x) = x \sin \frac{1}{x}$ and $g(x) = x^2 \sin \frac{1}{x}$ for $x \neq 0$, and $f(0) = g(0) = 0$. Show that both are continuous at 0, that f is not differentiable at 0, and that g is differentiable at 0 with $g'(0) = 0$.

(5) **Derivatives and tangent lines.**
   (a) Differentiate the following functions (wherever they are differentiable):

   (i) $3x\sqrt[3]{1 + x^2}$
   (ii) $x \ln x$
   (iii) $\cos(e^{x^2+1})$
   (iv) $-\dfrac{1}{(4x - 1)^3}$
   (v) $x \tan(x^3)$

   (b) Write the equation of the tangent line to the graph of f at x_0:

(i) $f(x) = \ln(3x-2)$, $x_0 = 2$
(ii) $f(x) = \dfrac{x}{1+x^2}$, $x_0 = 1$
(iii) $f(x) = e^{\sqrt{2x+1}}$, $x_0 = 0$
(iv) $f(x) = \sin\frac{1}{x}$, $x_0 = \frac{1}{\pi}$

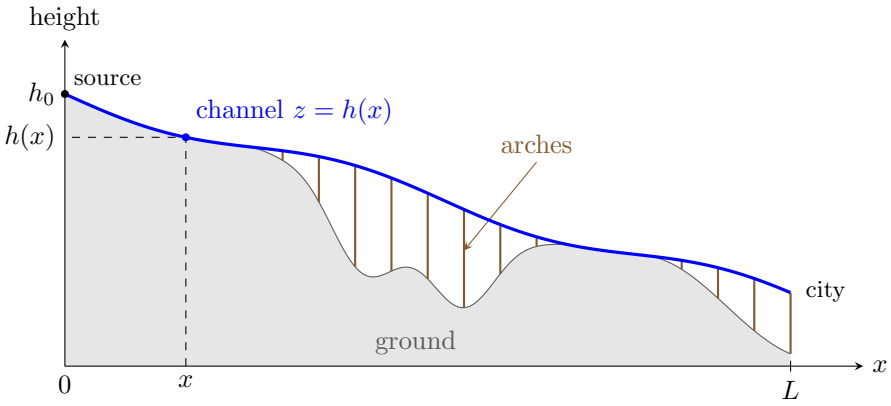
(6) **Linear approximation.** Recall that if f is differentiable at x_0, then $f(x_0+\Delta x) = f(x_0) + f'(x_0)\Delta x + o(\Delta x)$ as $\Delta x \to 0$.
(a) Use this with $f(x) = \sqrt{x}$ and $x_0 = 4$ to estimate $\sqrt{4.1}$. Compare with the value given by a calculator.
(b) The period of a pendulum of length ℓ is $T(\ell) = 2\pi\sqrt{\ell/g}$. Compute $T'(\ell)$ and show that, for a small change $\Delta\ell$ of the length, the relative change of the period is approximately half the relative change of the length: $\frac{\Delta T}{T} \approx \frac{1}{2}\frac{\Delta\ell}{\ell}$.
(c) A pendulum clock keeps exact time at 20°C. Its pendulum is a steel rod: if its length is ℓ at 20°C, then at $20°\text{C}+\Delta\theta$ it is $\ell(1+\alpha\Delta\theta)$, with $\alpha \approx 1.2\cdot10^{-5}$ per °C. Approximately how many seconds per day does the clock lose at 30°C?

Bonus problems

(B1) **Aqueducts.** Water in an aqueduct flows by gravity alone, so the channel has to descend all the way from the source to the city. Put the source at $x = 0$ and let $x \in [0, L]$ be the *horizontal* distance from the source, and let $h(x)$ be the height of the bottom of the channel above sea level; all lengths are in meters. We use the model

$$h(x) = h_0 - \alpha x - \beta \sin\left(\frac{x}{\ell}\right), \qquad x \in [0, L],$$

where $h_0, \alpha, \beta, \ell > 0$ are constants with $\frac{\beta}{\ell} < \alpha$. The term $-\alpha x$ describes the average descent; the sine term models a slope that varies along the route.



(a) Using the inequality $|\sin u - \sin w| \le |u - w|$ (Section 9.4), show that for all $0 \le x_1 < x_2 \le L$

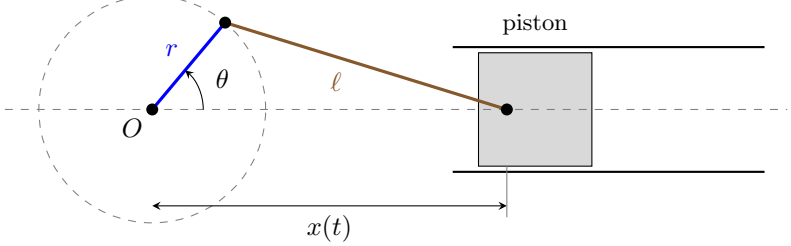
$$\left(\alpha - \frac{\beta}{\ell}\right)(x_2 - x_1) \;\le\; h(x_1) - h(x_2) \;\le\; \left(\alpha + \frac{\beta}{\ell}\right)(x_2 - x_1).$$

Deduce that h is strictly decreasing and Lipschitz on $[0, L]$. Conclude that h has an inverse $h^{-1} : [h(L), h(0)] \to [0, L]$, and that h^{-1} is continuous. What is the meaning of $h^{-1}(z)$ for the aqueduct?
(b) Compute $h'(x)$ directly from the definition of the derivative (*Hint:* proceed as in the computation of the derivative of $\sin x$ in the notes), and check that $\alpha - \frac{\beta}{\ell} \le -h'(x) \le \alpha + \frac{\beta}{\ell}$ for every $x \in (0, L)$.
(c) A leaf floats on the water. It leaves the source at time $t = 0$ and moves with constant horizontal speed $v > 0$, so that its height at time t is $z(t) = h(vt)$. Compute $z'(t)$ (*Hint:* use the Chain Rule). For $\alpha = 2\cdot10^{-3}$, $\beta = 1\,\text{m}$, $\ell = 1000\,\text{m}$ and $v = 1\,\text{m/s}$, how fast is the leaf descending when its horizontal position is $500\pi\,\text{m}$?

Historical remark. The Parco degli Acquedotti in Rome contains the remains of several aqueducts. One of them, called *Aqua Claudia*, was completed in 52 AD. Its channel is about 69 km long and descends from about 320 m to about 67 m above sea level. So $L \approx 69\,\text{km}$ and $h_0 \approx 320\,\text{m}$. On average, the channel descends only about 3.7 m per kilometer.

(B2) **Piston velocity.** In a car engine, a crank of radius r turns at constant angular speed $\omega > 0$ around a point O and drives a piston through a connecting rod of length $\ell > r$ (see the figure). The piston moves along a straight line through O. If $\theta = \omega t$ is the angle of the crank at time t, the distance from O to the piston pin is

$$x(t) = r\cos\theta + \sqrt{\ell^2 - r^2\sin^2\theta}, \qquad \theta = \omega t.$$



(a) Explain why the square root is defined for every t.
(b) Use the Chain Rule to compute the velocity $v(t) = x'(t)$ of the piston.
(c) At which angles θ is the piston momentarily at rest? What is its velocity at $\theta = \frac{\pi}{2}$? Evaluate it for an engine running at 3000 revolutions per minute with $r = 4\,\text{cm}$.
(d) What does $v(t)$ become when the rod is very long compared with the crank, i.e. as $\ell \to +\infty$?